

Problem–Solution Fit

Date	6 July 2026
Team ID	
Project name	Smart Agricultural Production Optimization Engine (OptiCrop)
Maximum marks	5 marks

Section	Content
Problem	Farmers face low crop yield due to unpredictable weather, improper irrigation, soil nutrient imbalance, and lack of data-driven decision-making.
Target Users	Farmers, Agricultural Officers, Agronomists, Government Agriculture Departments.
Current Challenges	Manual farming decisions, water wastage, excessive fertilizer use, pest attacks, and low productivity.
Proposed Solution	Develop an AI-based crop optimization system that analyzes soil, weather, and crop data to provide personalized farming recommendations.
Key Benefits	Higher crop yield, efficient irrigation, reduced fertilizer usage, early disease prediction, and increased farmer income.
Expected Outcome	Smart, sustainable, and profitable farming through AI-powered recommendations